Trustworthy Pipeline Leak Localization

AN ACOUSTIC EMISSION APPROACH

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Trustworthy Pipeline Leak Localization Based on Acoustic Emission Event Monitoring

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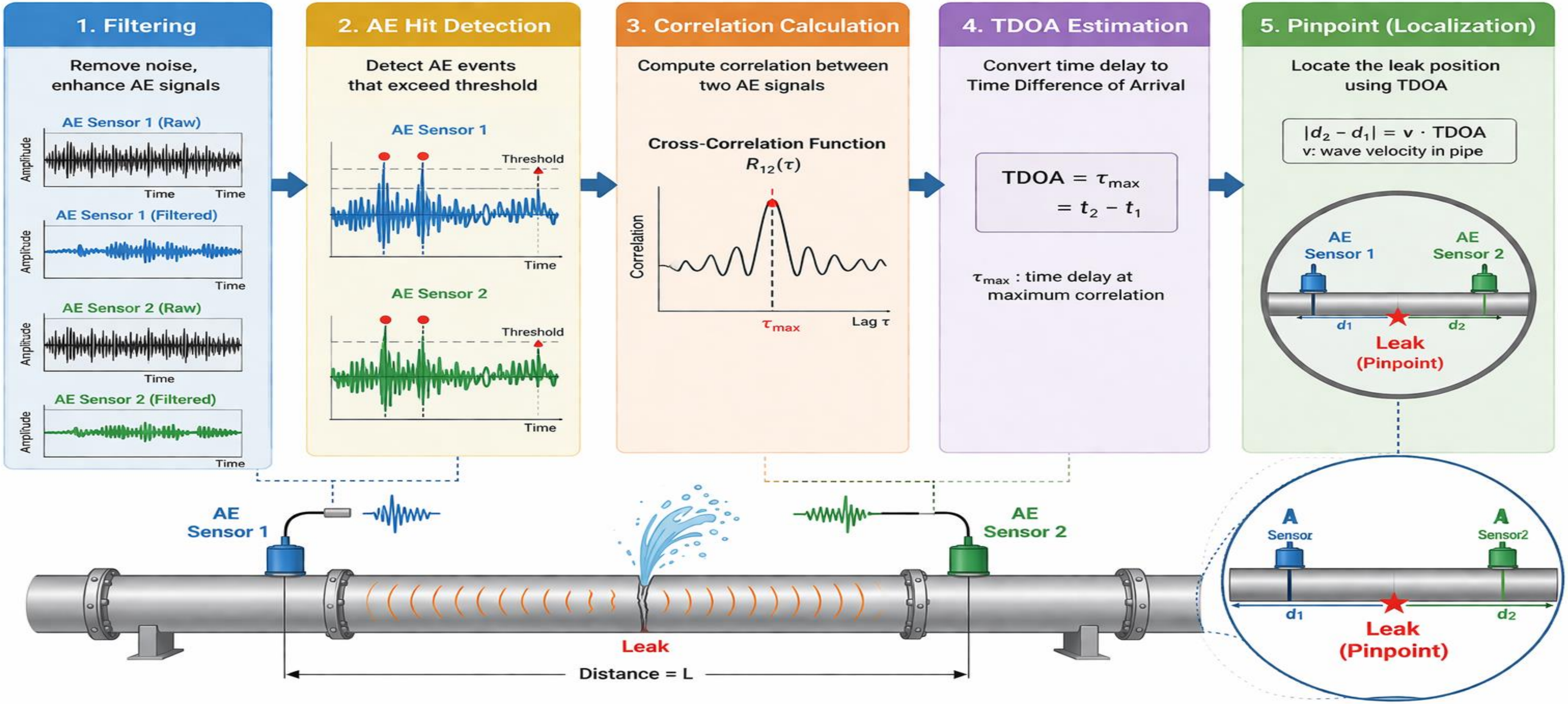
Introduction

- Pipelines are vital components of oil and gas transportation systems and of chemical and energy plants.
- One of the most prevalent issues faced by pipelines is leakage, which can occur due to corrosion, material fatigue, or external physical impacts.
- Pipeline leaks can result in consequences such as resource loss, environmental pollution, or adverse effects on community health.
- Identifying the location of pipeline leaks is a critical task to promptly detect the incident site for repair or replacement, ensuring the safe and uninterrupted operation of the system.
- Acoustic emission (AE) monitoring offers advantages over other methods due to its nondestructive, noninvasive, and highly sensitive nature.

Introduction



Introduction

- In this research,
  - A new scheme for AE event detection is proposed based on a novel test using instantaneous features including short-time energy (STE) and zero-crossing rate (ZCR).
  - An improved FIR filter developed from minimum entropy decomposition (MED) called IMED is proposed with optimized parameters.
  - A cross-correlation function called the cross-coherence function is proposed to enhance the accuracy and confidence of localization results.

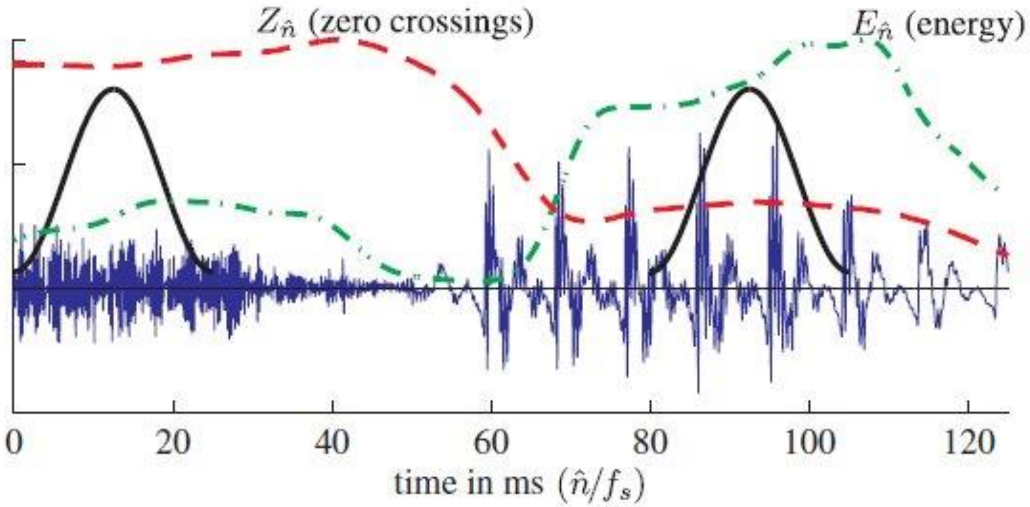
Background

- Instantaneous Features
- Zero-crossing rate:

$$Z[n] = \frac{1}{2L} \sum_{m=n-L+1}^{n} |\operatorname{sign}(x[m]) - \operatorname{sign}(x[m-1])|.$$

- Short-time energy:

$$E[n] = \sum_{m=n-L+1}^{n} x^2[m],$$



Background

- Minimum Entropy Deconvolution

We denote $\vec{x}=[x_1 \quad x_2 \quad \dots \quad x_N]^T$ as the input signal (to be filtered), $\vec{y}=[y_1 \quad y_2 \quad \dots \quad y_N]^T$ as the output signal (after filtering), and $\vec{f}=[f_1 \quad f_2 \quad \dots \quad f_K]^T$ as the impulse response of the FIR filter. The signal model of the causal discrete-time FIR filter is described as in (3.3), (3.4):

$$\vec{y} = \vec{f} * \vec{x} = X^T \vec{f} \tag{3.3}$$

$$X = \begin{bmatrix} x_1 & x_2 & x_3 & \dots & x_N \\ 0 & x_1 & x_2 & \dots & x_{N-1} \\ 0 & 0 & x_1 & \dots & x_{N-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & x_{N-K+1} \end{bmatrix}_{K\times N} \tag{3.4}$$

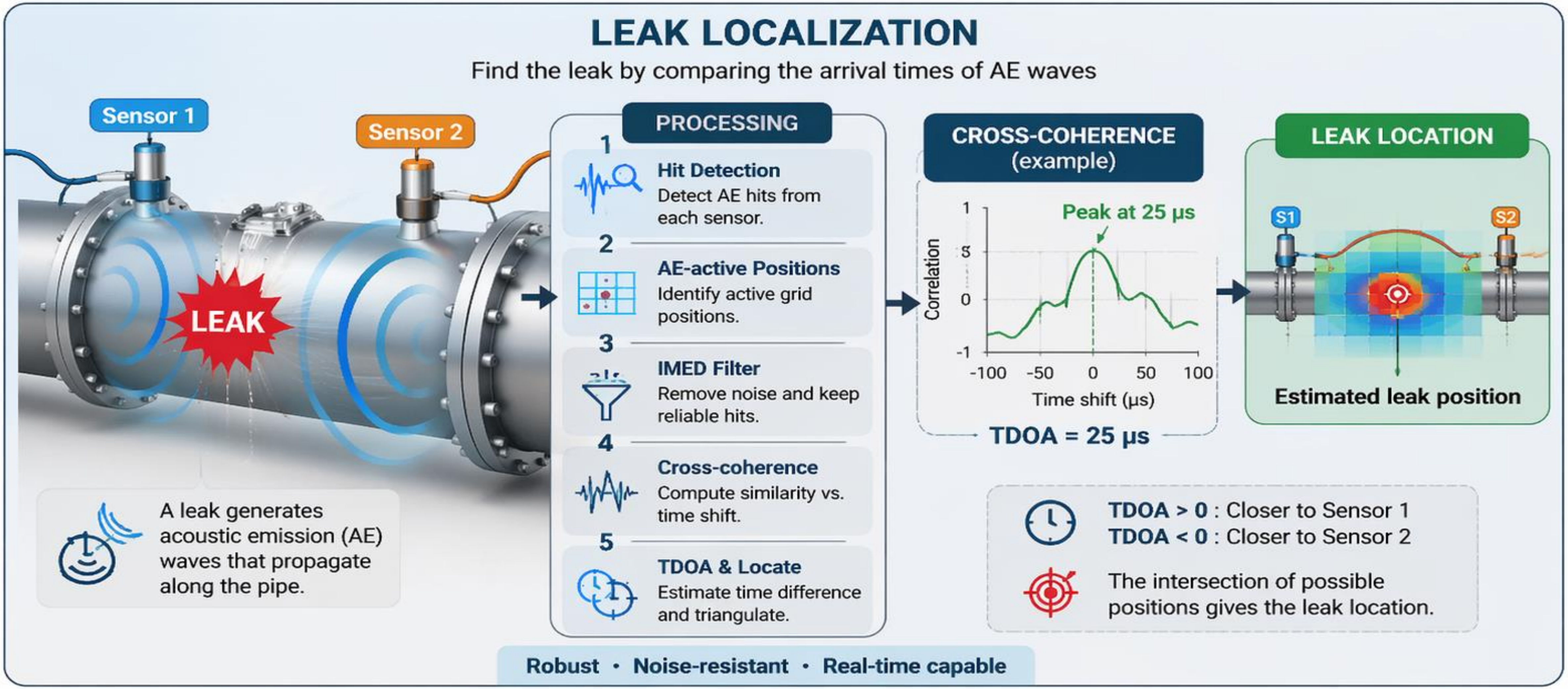
The objective function of MED is the kurtosis of the output signal, as shown in [52]:

$$\max_{\vec{f}} \frac{\sum_{n=1}^{N} y_n^4}{\left(\sum_{n=1}^{N} y_n^2\right)^2} \tag{3.5}$$

The iterative optimization process aims to determine the parameter set $\vec{f}$ of the FIR filter. After initiating with an initial value of $\vec{f}$, the output signal $\vec{y}$ can be computed iteratively. Subsequently, $\vec{f}$ is updated according to the update rule in [52]:

$$\vec{f} = \frac{\sum_{n=1}^{N} y_n^2}{\sum_{n=1}^{N} y_n^4} \left(XX^T\right)^{-1} X \left[y_1^3 \quad y_2^3 \quad \dots \quad y_N^3\right]^T \tag{3.6}$$

Methodology



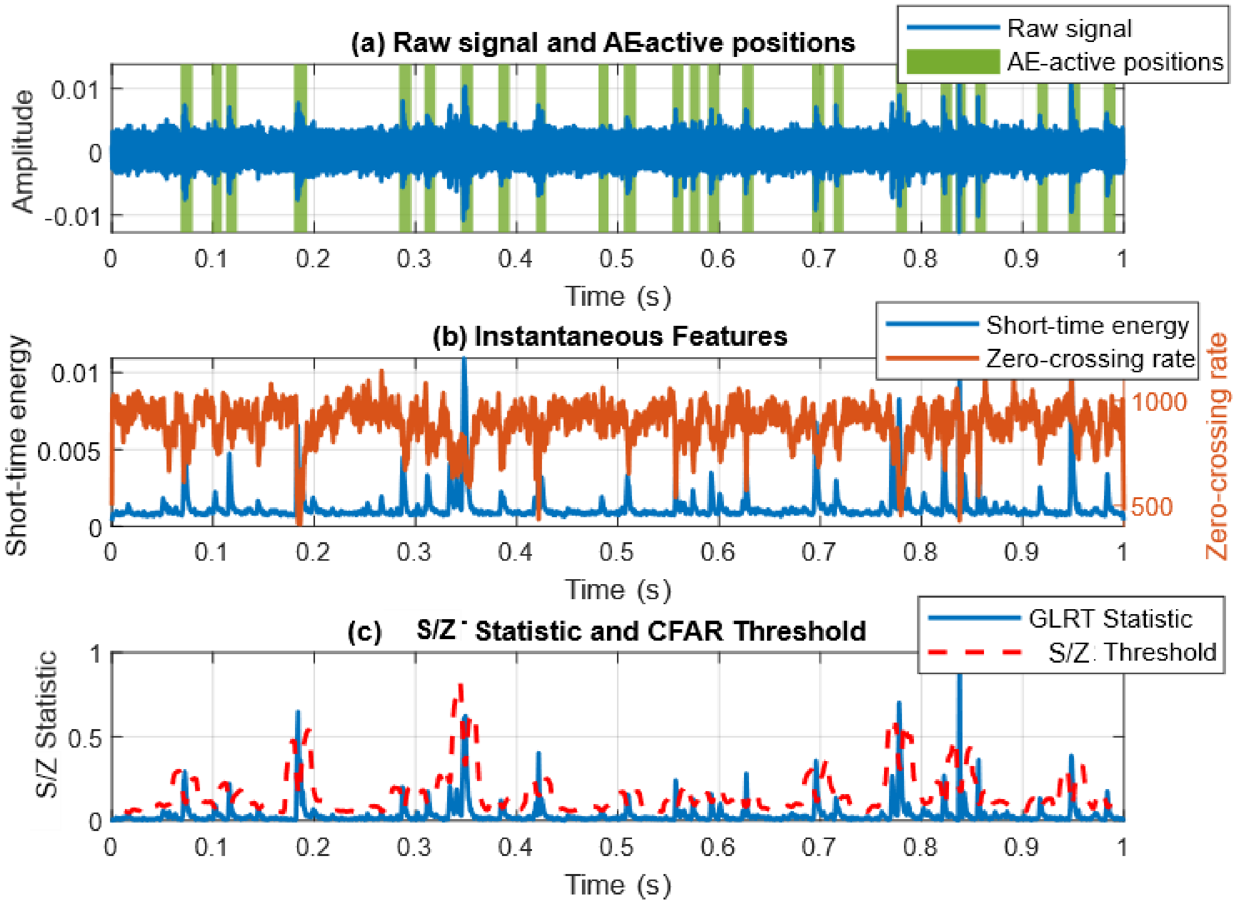
Methodology

- Hit Detection Rule:

$$\Lambda = \frac{STE}{ZCR} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma$$

- Adaptive Threshold:

$$\gamma = P_n \left(P_{fa}^{-1/n} - 1 \right)$$



Methodology

- IMED Filter
- Objective function:

$$\max_{\vec{f}} AD\_Norm = \frac{\vec{p}^T \vec{y}}{\left\| \vec{y} \right\|}$$

where **p** is the attention vector, calculated by convolving the positioning vector with the Hanning window function. The positioning vector indicates the AE-active positions, i.e., is equal to 1 at AE-positive positions and to 0 elsewhere.

Methodology

- IMED Filter
- Solution: To solve this problem, a simple approach is to calculate the derivative with respect to the filter coefficients and set it equal to the zero vector as follows:

$$\frac{d}{d\vec{f}} \frac{\vec{p}^T \vec{y}}{\|\vec{y}\|} = \sum_{i=1}^{N} \frac{d}{d\vec{f}} \frac{p_i y_i}{\|\vec{y}\|} = \vec{0} \qquad (3.10)$$

Using the formula for the derivative of a composite function, (3.10) becomes:

$$\frac{d}{d\vec{f}} \frac{\vec{p}^T \vec{y}}{\|\vec{y}\|} = \sum_{i=1}^{N} \|\vec{y}\|^{-1} p_i \vec{X}_i - \|\vec{y}\|^{-3} p_i y_i X\vec{y} = \vec{0} \qquad (3.11)$$

Methodology

- IMED Filter
- Solution:

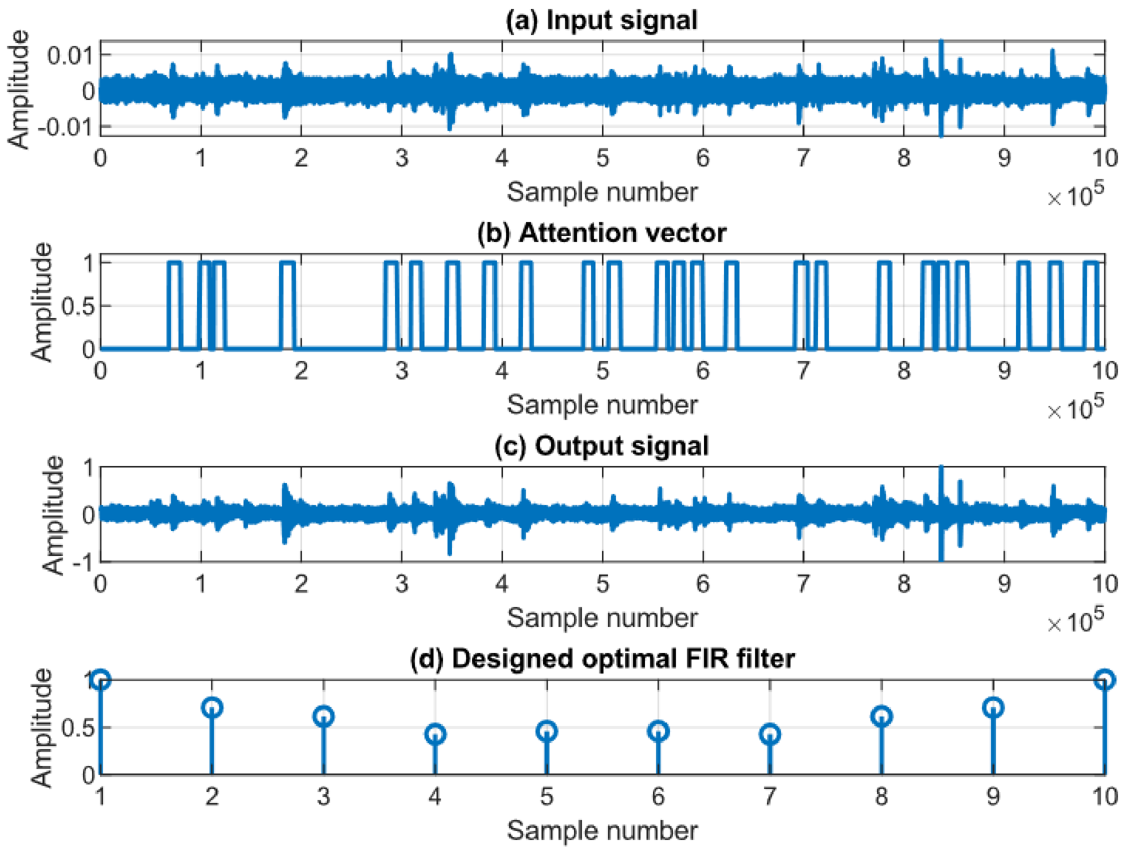
Let $G(\vec{f}) = \|\vec{y}\|^{-2} \vec{p}^T \vec{y}$, then we have $G(k\vec{f}) = k^{-1}G(\vec{f})$ for any constant k, since $\vec{y} = X^T \vec{f}$. Assume that $\vec{f}'$ is a solution of (3.13), then we have $G(\vec{f}')\vec{f}' = (XX^T)^{-1} X\vec{p}$. Considering $\vec{f} = G(\vec{f}')\vec{f}'$, the left-hand side of (3.13) is transformed into:

$$G(\vec{f})\vec{f} = G(\vec{f}')^{-1} G(\vec{f}')G(\vec{f}')\vec{f}' = G(\vec{f}')\vec{f}' = (XX^T)^{-1} X\vec{p} \qquad (3.14)$$

Therefore, $\vec{f} = G(\vec{f}')\vec{f}'$ is also a solution of (3.13), meaning that $\vec{f} = (XX^T)^{-1} X\vec{p}$ is the desired filter.

Methodology

- IMED Filter
- Solution:



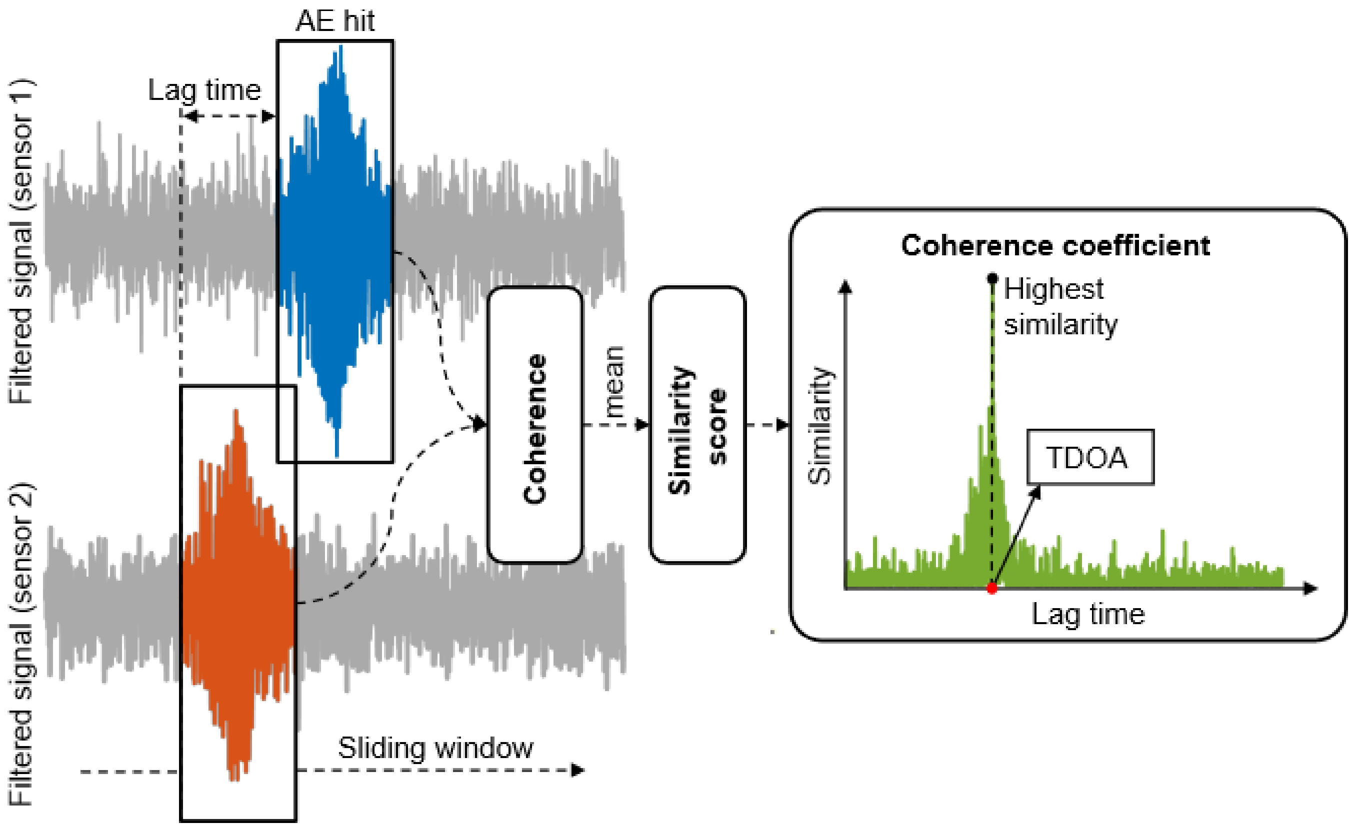
Methodology

- Cross-coherence function:

$$C_{xy}(f) = \frac{\left|P_{xy}(f)\right|^2}{P_{xx}(f)P_{yy}(f)}$$

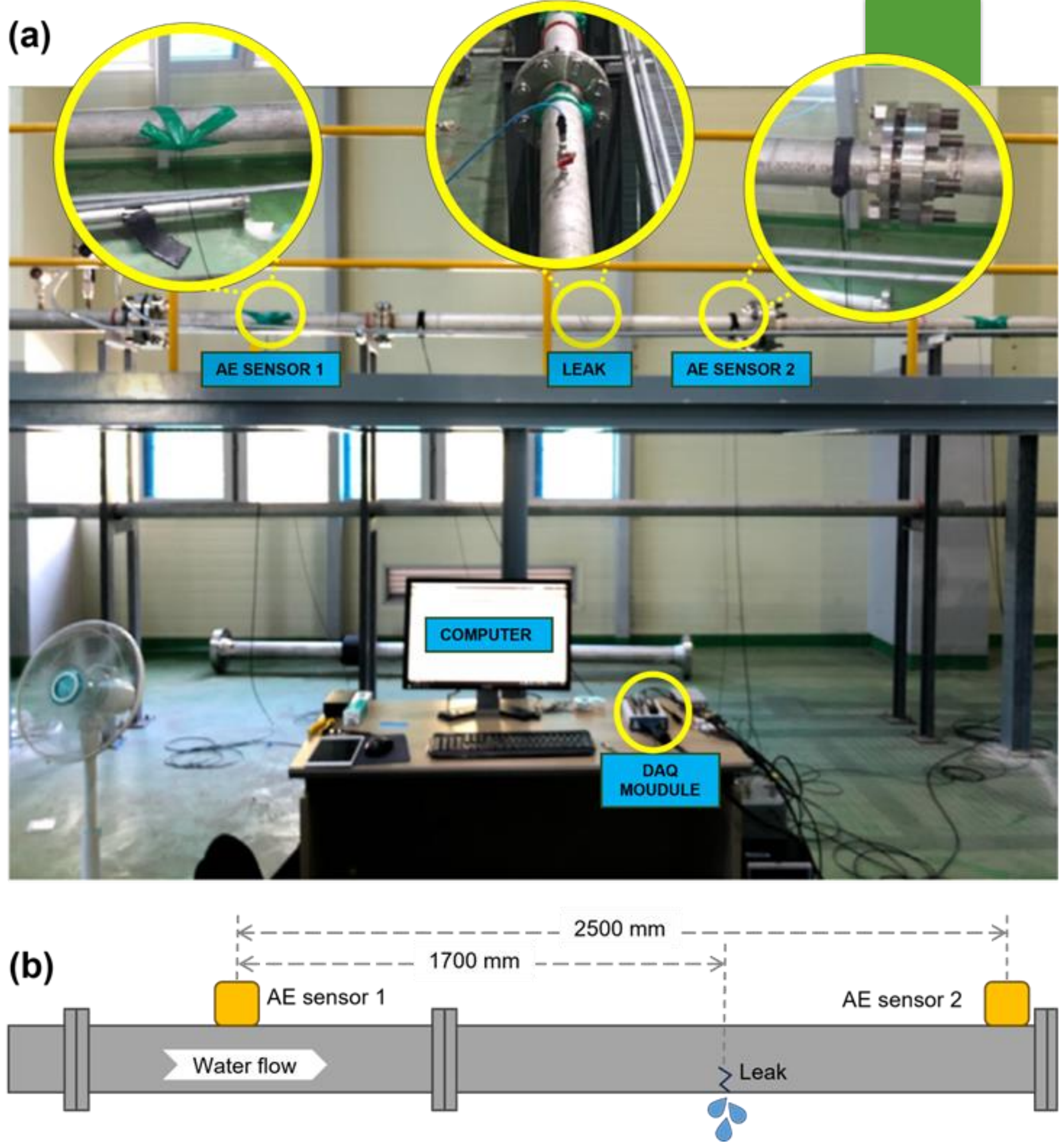
TDOA (Δt)

$$x = \frac{D - v\Delta t}{2}$$

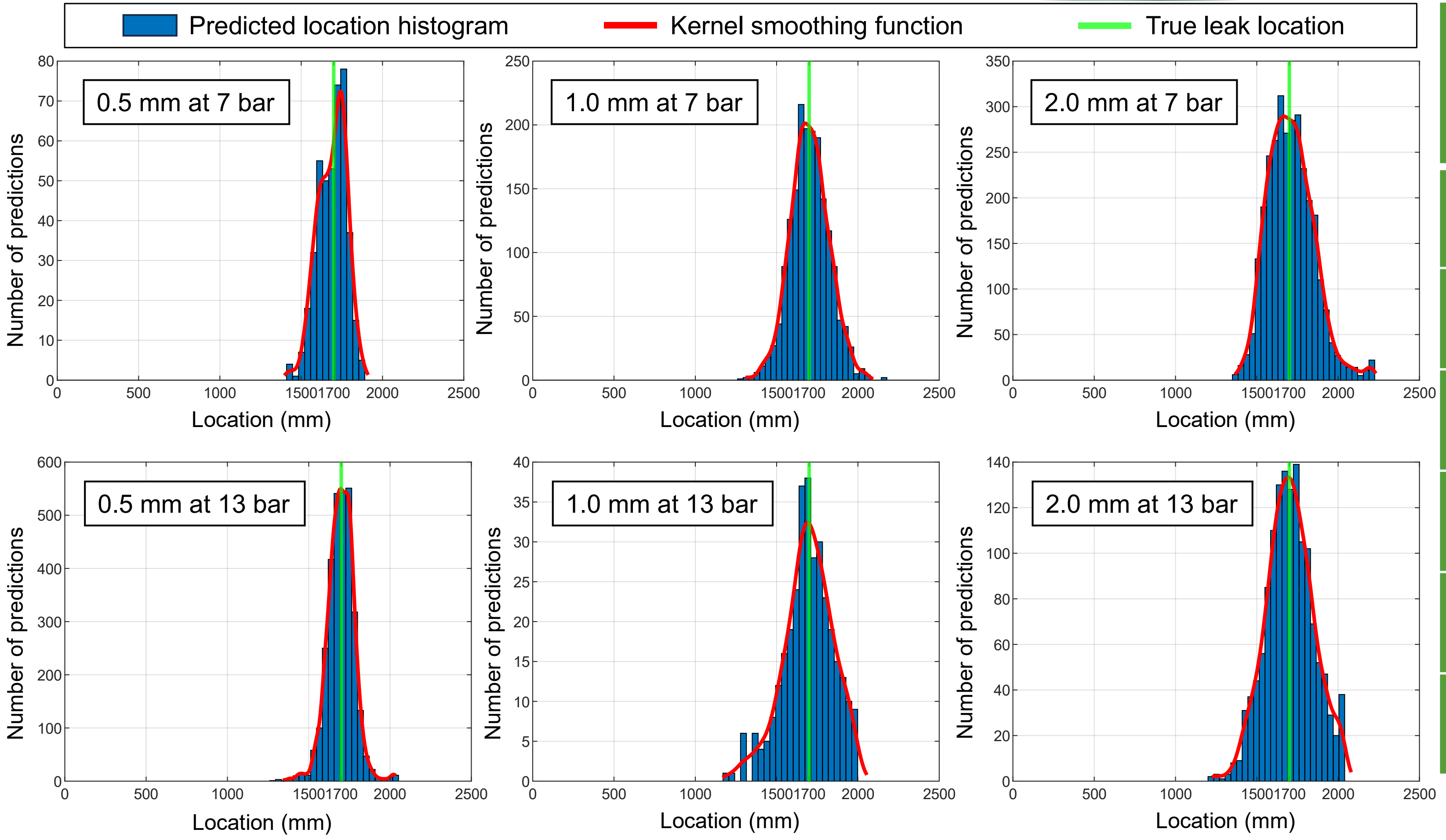


Results

- Data acquisition system:
- Leak size: 0.5, 1.0, 2.0
- Pressure: 7, 13 bar

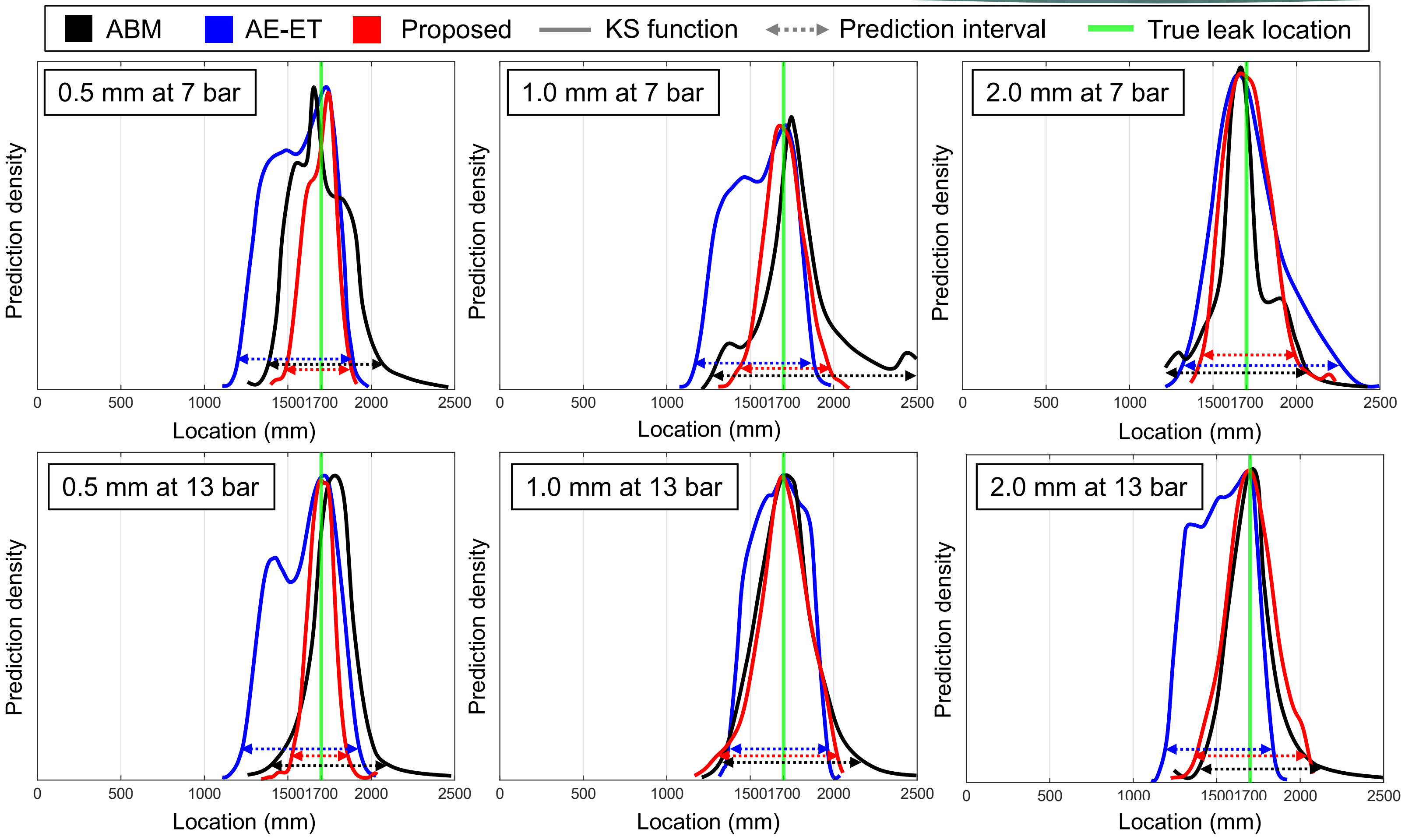


Results



Dataset	CCF	GCC	WPT-CCF [58]	TFD-CCF [59]	EMD-CCF [38]	ABM [9]	AE-ET [33]	Proposed method
0.5 mm at 7 bar	1650 ; 2.0	1561 ; 5.6	1556 ; 5.8	1560 ; 5.6	1555 ; 5.8	1750 ; 2.0	1673 ; 1.1	1674 ; 1.0
0.5 mm at 13 bar	1750 ; 2.0	1565 ; 5.4	1567 ; 5.3	1550 ; 6.0	1566 ; 5.4	1800 ; 4.0	1700 ; 0.0	1700 ; 0.0
1.0 mm at 7 bar	1800 ; 4.0	1564 ; 5.4	1568 ; 5.3	1560 ; 5.6	1565 ; 5.4	1700 ; 0.0	1700 ; 0.0	1695 ; 0.2
1.0 mm at 13 bar	1500 ; 8.0	1582 ; 4.7	1572 ; 5.1	1580 ; 4.8	1574 ; 5.0	1750 ; 2.0	1700 ; 0.0	1700 ; 0.0
2.0 mm at 7 bar	1200 ; 20.0	1565 ; 5.4	1564 ; 5.4	1555 ; 5.8	1562 ; 5.5	1650 ; 2.0	1666 ; 1.4	1684 ; 0.6
2.0 mm at 13 bar	1275 ; 17.0	1552 ; 5.9	1560 ; 5.6	1566 ; 5.4	1562 ; 5.5	1800 ; 4.0	1700 ; 0.0	1700 ; 0.0

Results



Data set (leak size at pressure)	CCF	GCC	WPT-CCF [58]	CCF-TFD [59]	EMD-CCF [38]	ABM [9]	AE-ET [9]	Proposed method
0.5 mm at 7 bar	1022	985	1190	1071	852	676	668	374
0.5 mm at 13 bar	942	721	1042	784	1270	690	672	315
1.0 mm at 7 bar	1108	1009	1156	919	851	1186	702	548
1.0 mm at 13 bar	1069	1167	1086	1049	896	803	604	687
2.0 mm at 7 bar	1337	981	902	1022	1099	856	910	557
2.0 mm at 13 bar	1254	1121	1246	1115	898	712	632	632

Conclusion

- This study proposed a new method with high confidence for the pipeline leak localization problem using AE monitoring
- Compared to existing methods, the proposed method significantly outperformed in both accuracy and confidence through rational improvements based on scientific foundations and rigorous reasoning.
- The results obtained promise the potential for real-world deployment in the industry.

THANK YOU FOR YOUR ATTENTION!

Q&A